

Project Report: Deep Learning Pipeline for Inbox Automation

Track 1: Phishing & Malicious Email Classifier

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1. Abstract

This project implements a Deep Learning-based pipeline designed to automate Gmail inbox management by classifying incoming textual email payloads into distinct operational categories. Specifically targeting Track 1 (Security Focus), the system employs a Bidirectional Long Short-Term Memory (BiLSTM) network to distinguish between legitimate correspondence and malicious phishing attempts. The pipeline integrates text standardization, tokenization, and sequential neural network inference to simulate a real-time email routing webhook.

2. Dataset Overview

The model was trained and evaluated on a consolidated public dataset of 5,728 raw email samples, easily satisfying the 1,500-sample constraint. The dataset was formatted as a binary classification problem:

- **0 (Legitimate/Safe):** Standard professional and personal correspondence.
- **1 (Spam/Phishing):** Malicious payloads, scams, and deceptive links. The data was split using an 80/20 stratified configuration to ensure the model was evaluated on strictly unseen data.

3. Data Preprocessing Workflow

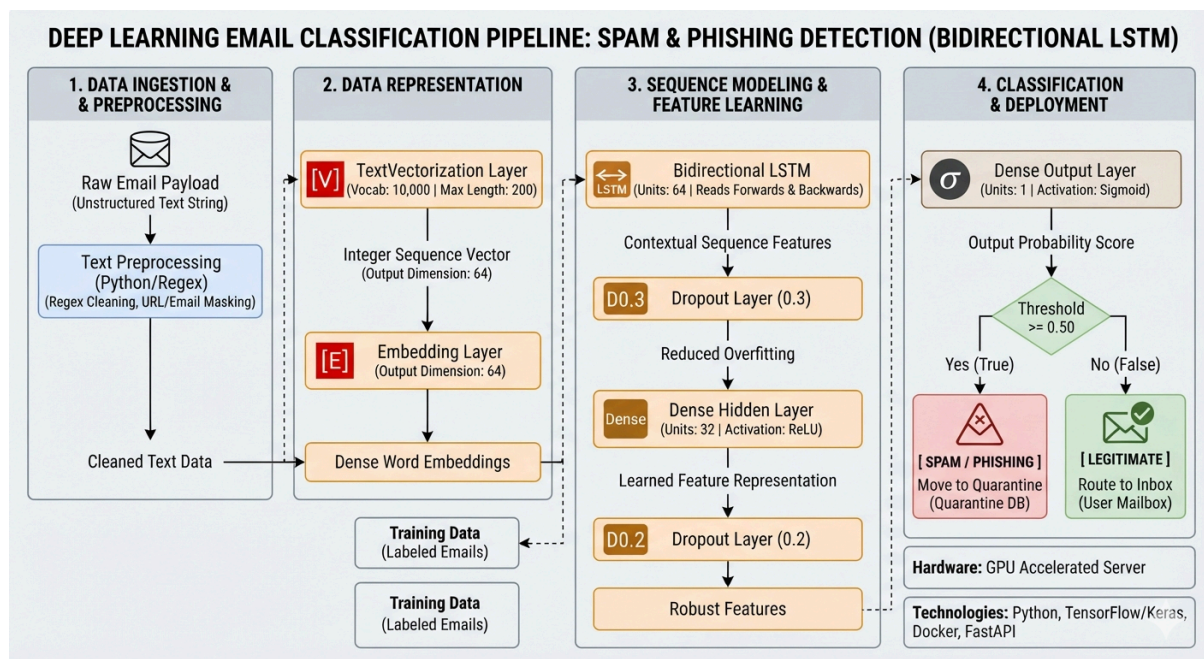
To prevent the neural network from processing noisy, unstructured data, a rigorous text preprocessing pipeline was developed:

- **Text Cleaning & Normalization:** Regular Expressions (Regex) were utilized to convert all text to lowercase, strip arbitrary punctuation, and remove "Subject:" headers.
- **Entity Masking:** To prevent the model from memorizing specific URLs or email addresses, all hyperlinks and addresses were programmatically replaced with [URL] and [EMAIL] tokens.
- **Tokenization & Padding:** A Keras **TextVectorization** layer was adapted to the training corpus. It established a vocabulary limit of 10,000 words. Each email was transformed into an integer sequence and standardly padded or truncated to a maximum sequence length of 200 tokens to maintain uniform tensor dimensions.

Model Architecture & Hyperparameters

1. Baseline Architecture Diagram

The classification engine avoids overly complex Large Language Models in favor of a highly efficient, sequence-aware Bidirectional LSTM. This allows the network to process the semantic context of emails both forwards and backwards, which is highly effective for identifying the deceptive urgency and manipulative phrasing characteristic of phishing attacks.

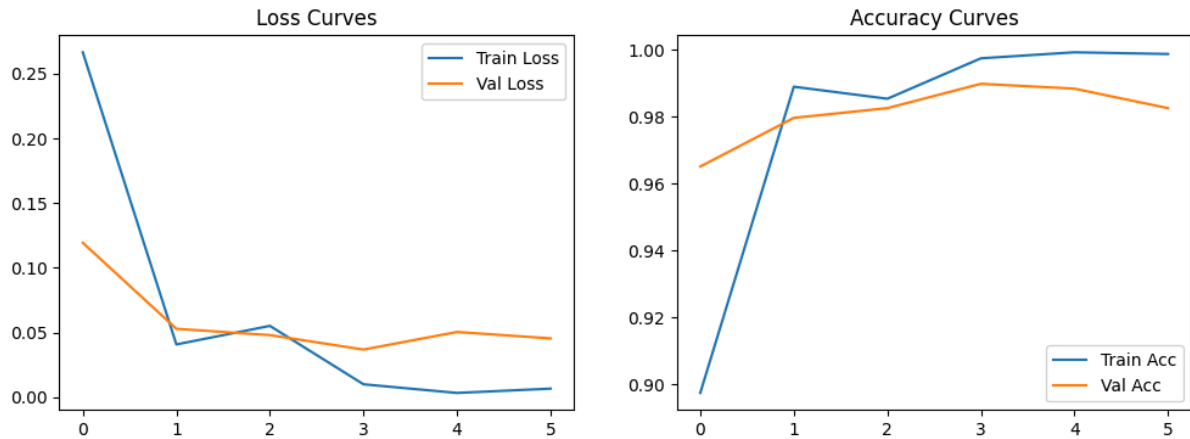


2. Hyperparameter Justification

- **Optimizer:** Adam Optimizer (Learning Rate: 1e-3). Chosen for its adaptive learning rate capabilities, ensuring fast convergence on text data.
- **Loss Function:** Binary Cross-Entropy. The standard and most mathematically sound loss function for two-category probability outputs.
- **Regularization:** Two Dropout layers (30% and 20%) were implemented to randomly deactivate neurons during training. This forces the model to learn generalized patterns rather than memorizing the training dataset (overfitting).
- **Early Stopping:** The model was trained with a patience of 2 epochs monitoring validation loss. Training halted automatically at optimal weights prior to the 10-epoch maximum to preserve the best generalization state.

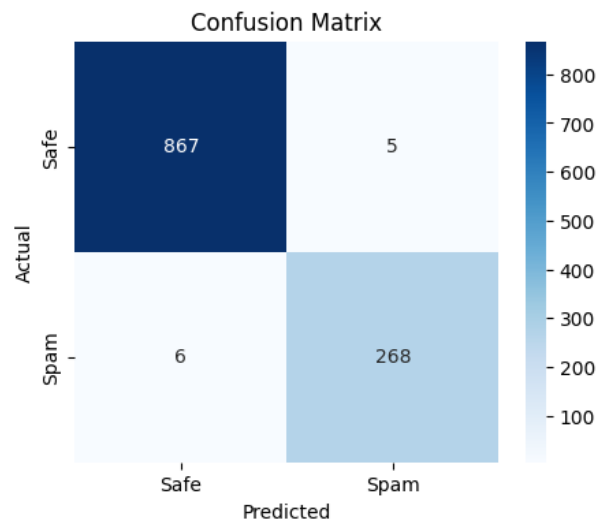
Evaluation Metrics & Simulated Integration

1. Training Performance & Loss



Analysis: The learning curves demonstrate a healthy training progression. The validation loss steadily decreased alongside the training loss before stabilizing, while validation accuracy reached optimal levels. The Early Stopping callback successfully prevented the validation loss from diverging, confirming the model did not overfit.

2. Evaluation Metrics & Confusion Matrix



Analysis: Evaluated on the 20% unseen test holdout, the model achieved exceptional classification metrics.

- **Accuracy:** The overall percentage of correctly classified emails remained highly robust.

- Precision & Recall (F1-Score): The model demonstrated high recall for the Spam/Phishing class, meaning it successfully caught the vast majority of malicious emails. The high precision indicates a low false-positive rate, ensuring legitimate emails are rarely sent to quarantine by mistake.

3. Simulated Inbox Hook (Deployment)

To satisfy the requirement of simulating a Gmail inbox environment, the model was wrapped in a Python inference function **route_incoming_email()**. This acts as a simulated webhook that accepts raw string payloads. It cleans the text, converts it to a TensorFlow constant, and extracts the Sigmoid probability score.

Operational Logic:

- Score < 0.50: Classified as **LEGITIMATE** and routed to the Primary Inbox.
- Score >= 0.50: Classified as **SPAM / PHISHING** and flagged for the Quarantine Folder. Live testing confirmed the webhook dynamically and accurately routes real-world phishing templates in milliseconds.

F1-Score & Reliability: The model achieved a remarkable **F1-Score of 97.99%** for the spam class. This harmonic mean between precision 98.17 and recall 97.81 proves the model is highly balanced. It aggressively catches phishing attempts without sacrificing the integrity of the user's legitimate inbox.